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pratham
Technology First

AGENTIC AI ENGINEERING

TRAINING PROGRAM





AGENTIC AI ENGINEERING

Agentic AI Engineering Training is an end-to-end program covering Python, Data Science, Machine Learning, Deep Learning, NLP, Generative AI, LLMs, Agentic AI, RAG, Multi-Agent Systems, AI protocols, and production deployment.

Tools & Frameworks



LangChain

Framework for building LLM and agent applications.



LangGraph

Framework for stateful and multi-step agent workflows.



OPENAI AGENTS SDK

Build agents with tools, guardrails, tracing, and handoffs.



Framework for collaborative multi-agent systems.



Toolkit for developing and deploying AI agents.



STRANDS AGENTS SDK

Build agents with AWS and Bedrock integration.



Azure AI Foundry

Platform for developing and deploying AI solutions.



Semantic Kernel

Framework for integrating AI models, tools, and workflows.



LlamaIndex

Build data-connected agents and retrieval workflows.



MCP

Protocol for connecting AI agents with tools and external data.



A2A

Protocol for communication between AI agents.



n8n

Platform for AI-powered workflow automation.



CAREER OPPORTUNITY

**Agentic AI
Engineer**

**AI Agent
Engineer**

**Generative AI
Engineer**

**AI/ML
Engineer**

**LLM
Engineer**

**AI Application
Developer**

**Machine
Learning
Engineer**

**AI Solutions
Architect**

RAG Engineer

**Multi-Agent
Systems
Engineer**

**AI Automation
Engineer**

**AI Workflow
Engineer**

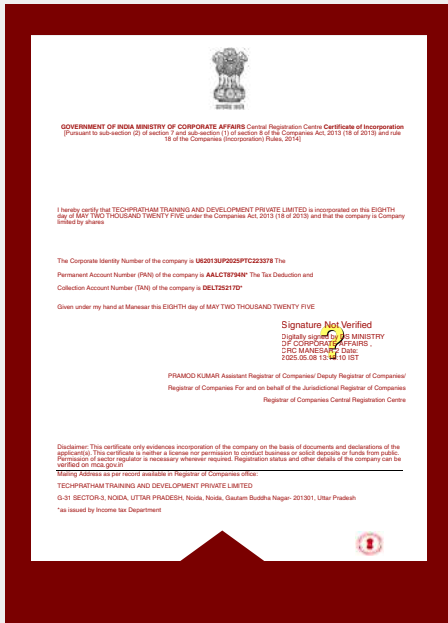
**NLP
Engineer**

**AI Software
Engineer**

**GenAI
Application
Developer**

**AI Platform
Engineer**

WE ARE AFFILIATED WITH DIFFERENT PARTNERS



ABOUT US

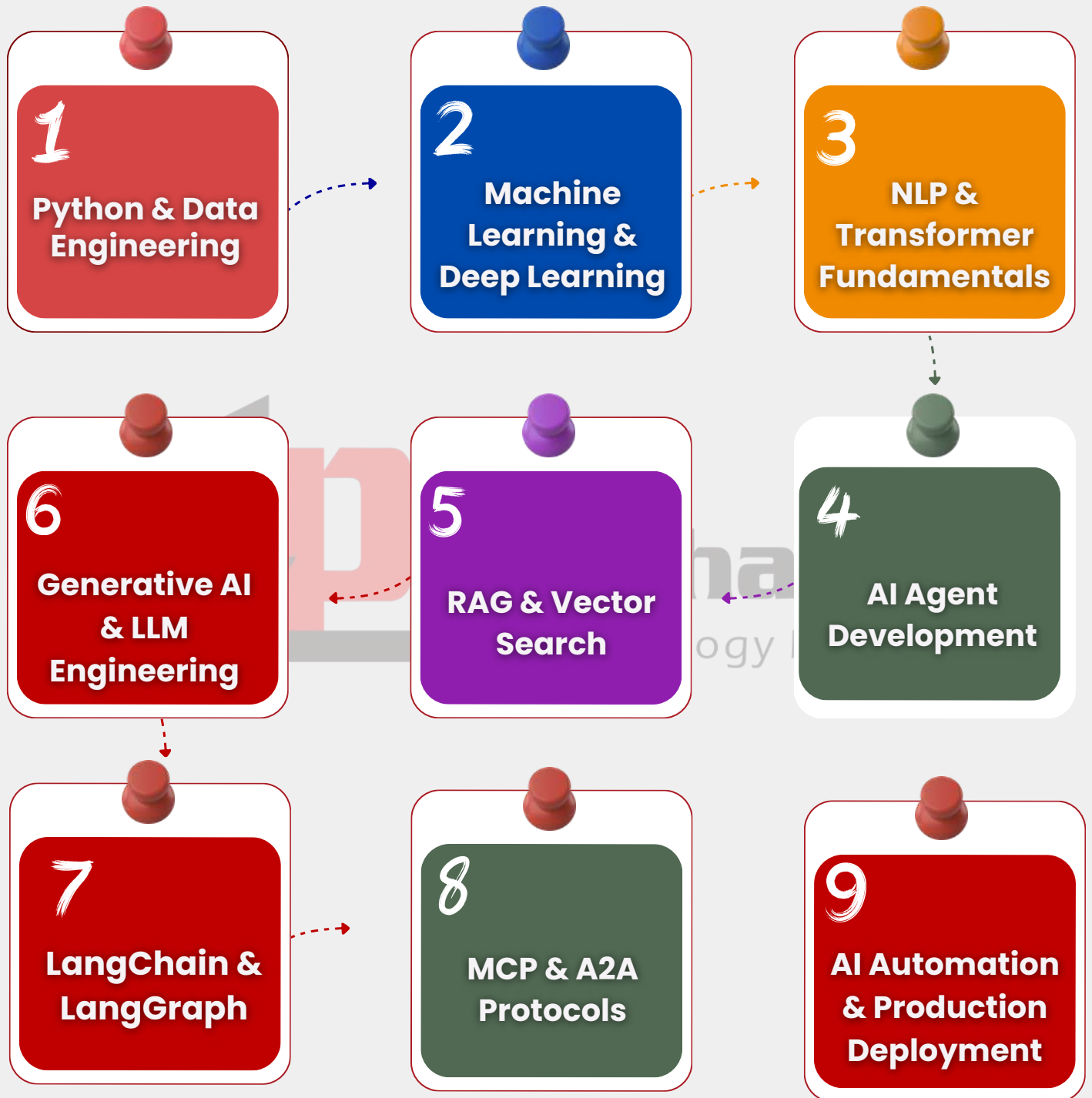
At **TechPratham**, we are committed to empowering careers through industry focused, practical training across multiple sectors. Our programs are carefully designed to make you job ready through hands-on learning, real time projects, and expert mentorship.

Learners benefit from extended platform access, 24/7 support, recorded sessions, and a truly hassle free learning experience. All our trainings are delivered by seasoned industry professionals with over a decade of real world expertise, ensuring deep insights and practical exposure. Backed by a dedicated placement team offering 100% job support, we continue to guide you until you achieve your ideal career milestone. With specialized departments and round-the-clock assistance, **TechPratham** stands as your trusted partner for skill excellence and career growth across diverse industries.





TOP SKILLS & TOOLS



OUR PLACEMENT PROCESS



PART 01

PROGRAMMING & DATA FOUNDATIONS

MODULE 1: PYTHON PROGRAMMING FOUNDATIONS

- Python Syntax, Data Structures & OOP
- File Handling & Working with APIs
- Virtual Environments & Package Management
- Writing Modular, Production-Grade Python

MODULE 2: NUMPY & PANDAS – DATA HANDLING

- NumPy Arrays & Vectorized Operations
- Pandas DataFrames – Loading, Cleaning, Transforming
- Exploratory Data Analysis (EDA)
- Data Visualization Basics (Matplotlib/Seaborn)

PART 02

AI, ML, DL & NLP FOUNDATIONS

MODULE 3: MACHINE LEARNING OVERVIEW

- Supervised vs Unsupervised Learning
- Regression & Classification Algorithms
- Model Evaluation & Metrics

- Feature Engineering & Preprocessing
- Scikit-learn Workflow

MODULE 4: DEEP LEARNING OVERVIEW

- Neural Network Fundamentals
- Training, Loss Functions & Optimizers
- CNNs & RNNs (Conceptual Overview)
- Intro to PyTorch / TensorFlow
- Transfer Learning Basics

MODULE 5: NLP OVERVIEW

- Text Preprocessing & Tokenization
- Word Embeddings (Word2Vec, GloVe)
- Transformer Architecture Basics
- Intro to Pretrained Language Models
- Bridge: From NLP to LLMs

PART 03

GENERATIVE AI & LLM FOUNDATIONS

MODULE 6: FOUNDATIONS OF GENERATIVE AI

- Generative vs Discriminative Models
- Model Types: Base, Instruct, Chat, Multimodal, Reasoning
- Open-Source vs Closed / Proprietary Models
- Model Selection Trade-offs: Cost, Latency, Context Length

MODULE 7: PROMPTING & LLM APIS

- Prompt Engineering: Zero-shot, Few-shot, Chain-of-Thought
- System, User & Assistant Roles
- Calling LLM APIs (OpenAI, Anthropic, Google, Hugging Face, Ollama)
- Streaming, Function/Tool Calling & Structured Outputs
- Rate Limits, Token Costs & API Best Practices

MODULE 8: EMBEDDINGS & VECTOR SEARCH

- What Embeddings Are & How They're Trained
- Embedding Models (OpenAI, Cohere, Sentence-Transformers)
- Vector Similarity & Distance Metrics
- Vector Databases (FAISS, Chroma, Pinecone, Qdrant)
- Intro to Retrieval-Augmented Generation (RAG)

PART 04

AGENTIC AI CORE FRAMEWORKS & PROTOCOLS

MODULE 9: INTRODUCTION TO AGENTIC AI

- What Are AI Agents?
- Agentic AI vs Generative AI vs AI Agents
- Multi-Agent Systems
- Agent Design Patterns

- Agentic Protocol Layer
- Agent Framework Landscape
- Reasoning Models in Agents

MODULE 10: PYDANTIC – DATA VALIDATION & STRUCTURED OUTPUTS

- Pydantic Fundamentals
- Data Validation & Handling
- Advanced Validation
- Pydantic in Agentic Pipelines
- Monitoring with Logfire

MODULE 11: LANGCHAIN & LCEL

- LangChain Core Components
- Data Ingestion
- Embeddings & Vector Stores
- LangChain Expression Language (LCEL)
- LangServe & Production APIs

MODULE 12: LANGGRAPH – STATEFUL AGENT WORKFLOWS

- Graph Concepts
- Agents & Memory
- Message Handling

- Human-in-the-Loop
- LangGraph Deployment

MODULE 13: MCP – MODEL CONTEXT PROTOCOL

- MCP Fundamentals
- Building MCP Servers
- MCP Ecosystem
- MCP with Agent Frameworks
- MCP in Production

MODULE 14: A2A – AGENT-TO-AGENT PROTOCOL

- A2A Fundamentals
- A2A Core Components
- Building A2A-Compliant Agents
- A2A in Practice

PART 05

RETRIEVAL, CONTEXT & MULTI-AGENT SYSTEMS

MODULE 15: RAG & AGENTIC RAG

- RAG Fundamentals
- Advanced Retrieval
- RAG Evaluation

- Adaptive RAG
- Agentic RAG
- GraphRAG

MODULE 16: CONTEXT ENGINEERING

- Prompt Engineering vs Context Engineering
- Structured System Prompts
- Memory Systems
- Context Window Management
- Dynamic Context Assembly
- Context Testing & Optimization

MODULE 17: OPENAI AGENTS SDK

- SDK Fundamentals
- Tools & Tool Patterns
- Multi-Agent Handoffs
- Guardrails
- Sessions & Tracing
- Realtime Voice Agents

MODULE 18: CREWAI – MULTI-AGENT COLLABORATION

- CrewAI Architecture
- CrewAI Flows

- Tools & Advanced Features
- Complex Multi-Agent Crews
- Integration & Monitoring

PART 06

PLATFORM & ECOSYSTEM INTEGRATIONS

MODULE 19: GOOGLE ADK – AGENT DEVELOPMENT KIT

- ADK Architecture
- Tools & MCP
- Model Flexibility
- A2A with ADK
- Evaluation & Deployment
- Observability

MODULE 20: AWS STRANDS AGENTS SDK

- Strands Architecture
- Agents & Tools
- MCP Integration
- Multi-Agent Patterns
- AWS & Bedrock Integration
- Deployment & Observability

MODULE 21: AZURE AI FOUNDRY & SEMANTIC KERNEL

- Semantic Kernel Fundamentals
- Agent Patterns
- Azure AI Foundry
- MCP & A2A Integration
- Azure Deployment
- Monitoring

MODULE 22: LLAMAINDEX – DATA-CENTRIC AGENTS

- LlamaIndex Fundamentals
- Agentic Workflows
- Agent Tools & Memory
- LlamaIndex & MCP
- Advanced Indexing

MODULE 23: CLAUDE CODE – AGENTIC CODING

- Claude Code Fundamentals
- Agentic Coding Workflows
- MCP with Claude Code
- Extended Thinking & Automation
- Multi-Agent Coding Pipelines

MODULE 24: N8N – AI WORKFLOW AUTOMATION

- n8n Fundamentals
- APIs & Webhooks
- Error Handling
- AI Agents in n8n
- AI Automation Projects
- Integrations & Production



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Bhaji Tyagi

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DIRECTOR & AUTHORIZED
SIGNATORY



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IT'S NECESSITY.**

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